OpenLCB Working Note	
DCC Detection Protocol	
Sep 28, 2026	Preliminary

1 Introduction

A Working Note is an intermediate step in the documentation process. It gathers together the content from various informal development documents, discussions, etc into a single place. One or more Working Notes form the basic for the next step, which is one or more Standard/TechNote pairs.

This working note describes a OpenLCB protocol for a detector to report its detection of a train, including the DCC address of an associated DCC decoder.

"Detector" refers to a fixed-location electronic device that can interact with a DCC decoder to ascertain the address of the DCC decoder while the decoder is in a specific area or at a specific point. Possible implementations include but are not limited to RailCom detectors, Digitrax detectors, RFID and QR readers.

1.1 Served Use Cases

A train enters a current-detected block. A detector recognizes this, successfully queries the decoder for its address, and reports that information on the OpenLCB bus. That information results in a signal controller sending a speed command such as Slow or Stop to that particular detector.

A train crosses an RFID reader which determines the train's DCC address and reports that address on the OpenLCB bus.

A train exits a current-detected block. A detector recognizes this, and reports that interaction on the OpenLCB bus.

A train enters a current detected block. A detector recognizes this, and attempts unsuccessfully to query the decoder for its address. The detector emits a report without decoder address information.

1.2 Unserved Use Cases

Being able to report information on a native OpenLCB (non-DCC) train is an unserved use case. This protocol doesn't provide a way to report the Unique ID of a fully OpenLCB train. All devices being reported need to have a DCC address.

Having a monitor tool that can show all detection information without configuration is a unserved use case. The Unique IDs of the detectors, hence the upper bytes of Event IDs to be consumed, are determined by a pre-configuration process in the consumer. They can't be determined automatically.

DRAFT

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.

2 Specified Sections

This is the usual section organization for a Technical Note, to accumulate the Standard and Technical Note content in its eventual order.

2.1 Introduction

Note that this section of the Standard is informative, not normative.

2.2 Intended Use

Note that this section of the Standard is informative, not normative.

This protocol is intended to provide a standard way to transfer train location and DCC decoder identification information on OpenLCB DCC layouts.

2.3 Reference and Context

This protocol builds on the Event Transport Protocol as defined by that Standard and Technical Note.

2.4 Message Formats

Detection events are produced via PCER messages with an Event ID which contains

1) 6 byte unique ID of the detector,
2) Two bits to represent entry/exit and ~~forward/reverse.~~orientation. These are coded:
   - 0b00 unoccupied, i.e., exit
   - 0b01 occupied, ~~forward~~facing West, ~~ie~~i.e., entry
   - 0b10 occupied, ~~reverse~~facing East, ~~ie~~i.e., entry
   - 0b11 occupied, ~~direction~~orientation unknown, ~~ie~~i.e., entry
3) 14 bits of DCC address
   A) If the high bits are 0x38 this represents a DCC short address
   B) If the high bits are 0x39 this represents a DCC consist address
   C) ~~An address of 0x3800 (short address 0) represents "unknown".~~An address of 0x3800 (short address 0) represents that the track is empty, i.e. there is no decoder present.
   D) Otherwise it's a DCC long address

The encoding of DCC addresses into 14 bits follows the S-9.2.1.1 and RCN-218 standards.

The 0x3800 value is used for ~~unknown~~track empty instead of 0x0000 because 0x0000 is long address zero, which is a valid (though uncommon) long address.

A detector may, but does not have to, use the 0x3800 "track is empty" by for example producing a "'track is empty' leaves" event when the track becomes occupied, and a "'track is empty arrives" when the track becomes unoccupied. For example:

DRAFT

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.

06.80.99.30.03.50.38.00 - "track is empty" leaves

06.80.99.30.03.50.F8.03 - 3S arrives

06.80.99.30.03.50.38.03 - 3S leaves

06.80.99.30.03.50.F8.00 - "track is empty" arrives

2.4.1 Orientation Definitions

Detectors that report orientation (such as RailCom detectors) determine the physical orientation of the decoder on the track, rather than the internal throttle direction ("forward" or "reverse") commanded to the locomotive motor. Following DCC standards, the two detectable track orientations are specified as "facing West" and "facing East".

The normative definitions of "West" and "East" are taken from RailCommunity standard RCN-217 (Bi-directional communication, Section 5.5, Endnote 5):

> „West“ ist definiert als bei positiver Polarität des Gleissignals nach [RCN-210] Abschnitt 2.3 vorwärts und rückwärts bei negativer Polarität. „Ost“ ist entsprechend umgekehrt definiert.
>
> (Translation: "West" is defined as forward movement with positive polarity of the track signal according to [RCN-210] Section 2.3, and reverse movement with negative polarity. "East" is correspondingly defined vice versa.)

RCN-210 Section 2.3 establishes the electrical and wiring polarity:

1) The red decoder wire connects to the right-hand rail (facing forward from the cab).
2) The black decoder wire connects to the left-hand rail.
3) "Positive polarity" is defined as the right rail being positive with respect to the left rail during the first half of the DCC bit cycle: V(right rail - left rail) > 0.

Standard Layout Reference:

1) For track running East-West, the North rail is wired as Rail A (positive polarity in the first half of the DCC bit), and the South rail is wired as Rail B. In a common depiction, this means that on the track in front of you leftwards is Westbound, rightwards is Eastbound, the back rail (the rail that is further away from you) is Rail A, wired with red wire.
2) A locomotive facing West has its right rail on the North rail (Rail A). In forward motion, its right rail is positive, satisfying the RCN-217 definition for "West". In reverse motion facing East, its right rail is Rail B (negative), which also evaluates to "West".
3) A locomotive facing East has its right rail on the South rail (Rail B). In forward motion, its right rail is negative, satisfying the RCN-217 definition for "East".

Translation to RailCom Detector Hardware:

1) During the RailCom cutout, RailCom detectors measure current flow between Rail A and Rail B:

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.

1. When the North rail (Rail A) sources RailCom current and the South rail (Rail B) sinks current—producing a positive voltage across the detector measurement resistor on (Rail A - Rail B)—the locomotive is detected as facing West (code 0b01).
2. When the South rail (Rail B) sources RailCom current and the North rail (Rail A) sinks current—producing a negative voltage on (Rail A - Rail B)—the locomotive is detected as facing East (code 0b10).

Unknown orientation needs to be used in a variety of situations:

- If a RailCom detector is wired on a three-rail layout (where direction of current is fixed / does not correlate with orientation).
- If a RailCom detector is not equipped with hardware for distinguishing positive / negative current flow direction.
- If the detector uses a technology (e.g. RFID) that represents DCC address but does not represent orientation.

Bit Mapping:

0b01 Occupied, facing West

0b10 Occupied, facing East

0b11 Occupied, orientation unknown (used when the detector can read the decoder address, but cannot determine track orientation)

Examples of Orientation Events:

For a detector with Node ID 06.80.99.30.03.50 detecting locomotive short address 3 (0x3803):

06.80.99.30.03.50.78.03 - Locomotive 3S arrives, facing West (0b01)

06.80.99.30.03.50.B8.03 - Locomotive 3S arrives, facing East (0b10)

06.80.99.30.03.50.F8.03 - Locomotive 3S arrives, orientation unknown (0b11)

06.80.99.30.03.50.38.03 - Locomotive 3S leaves (0b00)

2.5 States

This protocol has no states of its own.

An “exit Event ID can be sent at most once after a corresponding entry Event ID” state requirement was discussed but not accepted. Note that some hardware detectors determine only that a decoder has left the monitored section, not its address, so those scanners will be producing “NNN entry” followed by “(none) exit”.

2.6 [2.5] Interactions

A detector sends a PCER with an Event ID as described above when it detects entry of a decoder into the monitored location. It may also send a PCER when the exit of the decoder from the monitored location is detected.

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.

A detector, acting as a producer, must reply to Verify Events and Identify Producers messages with the appropriate event(s) or range.

Upon an Identify Producers message for an Event ID falling into the range:

- Detectors that do not produce Exit events shall reply with Producer Identified Unknown;
- Detectors that do not detect the orientation of a locomotive, upon receiving an Event ID with a definite orientation (East or West) shall reply with Producer Identified Unknown; otherwise the response shall be Valid or Invalid as follows:
  - A Locomotive Entry with Orientation Unknown Event ID shall receive the Producer Identified Valid response if and only if the locomotive is currently in the block.
  - A Locomotive Exit Event ID shall receive a Producer Identified Valid response if and only if the locomotive is currently NOT in the block.
  - A Locomotive Entry Orientation West Event ID shall receive a Producer Identified Valid response if and only if the locomotive is in the block with orientation West.
  - A Locomotive Entry Orientation East Event ID shall receive a Producer Identified Valid response if and only if the locomotive is in the block with orientation East.

In addition to the above, detectors that produce Exit events, upon receiving an Identify Producer message for the Event ID with suffix 0000 (Locomotive Exit Long Address 00), shall enumerate all currently registered locomotives that are inside the block and emit a Producer Identified Valid message for each of them according to the above rules; followed by a Producer Identifier Valid or Invalid message for event ID suffix 0000; followed by a Producer Range Identified message for the entire 64k Event ID range.

Consumers can consume the entire 16-bit range of the lower EventID bytes. Or they can consume just the specific EventIDs that correspond to the particular decoder(s) that they're tracking. In either case, they must respond to Verify Events and Identify Consumers messages with the appropriate event(s) or range.

There is no PIP bit allocated to this protocol.

3 Background Information

This protocol has been implemented in several products. It is known to work for its served use cases.

DRAFT

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.

160

Table of Contents

1 Introduction 1
1.1 Served Use Cases 1
1.2 Unserved Use Cases 1
2 Specified Sections 2
2.1 Introduction 2
2.2 Intended Use 2
2.3 Reference and Context 2
2.4 Message Formats 2
2.4.1 Orientation Definitions 3
2.5 States 4
2.6 Interactions 4
3 Background Information 5

DRAFT

Copyright 2011-~~2023~~2026. All rights reserved. This OpenLCB document is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0). See https://openlcb.org/licensing for more information.